

# SLM Benchmark Report

## 1. Executive Summary

This report compares the performance of Small Language Models (SLMs) for banking-related tasks. The benchmark covers latency, throughput, and semantic quality.

- Fastest Model: facebook/opt-125m (47.64 tokens/sec)
- Highest Quality: TinyLlama/TinyLlama-1.1B-Chat-v1.0 (Similarity: 0.72)

## 2. Detailed Analysis

The benchmark results highlight a distinct trade-off between inference speed and response quality. 'facebook/opt-125m' leads in throughput, generating 47.64 tokens per second. This speed is advantageous for real-time chat interfaces where low latency improves user satisfaction. However, it achieves a semantic similarity score of 0.37.

Conversely, 'TinyLlama/TinyLlama-1.1B-Chat-v1.0' demonstrates the highest understanding of banking context with a semantic similarity score of 0.72. This higher accuracy ensures that responses align closely with verified banking protocols, which is critical for compliance and trust. The trade-off is a lower throughput of 16.43 tokens per second.

Recommendation: For high-stakes queries (e.g., fraud reporting, fee disputes), prioritize the higher quality model. For general FAQs where speed is essential, the faster model may be sufficient.

## 3. Performance Data

| Model                              | Latency (s) | Tokens/s | Similarity |
|------------------------------------|-------------|----------|------------|
| Qwen/Qwen2-1.5B-Instruct           | 12.25       | 12.25    | 0.72       |
| Qwen/Qwen1.5-0.5B-Chat             | 5.00        | 30.04    | 0.71       |
| TinyLlama/TinyLlama-1.1B-Chat-v1.0 | 9.06        | 16.43    | 0.72       |
| HuggingFaceTB/SmolLM-1.7B-Instruct | 12.42       | 7.69     | 0.66       |

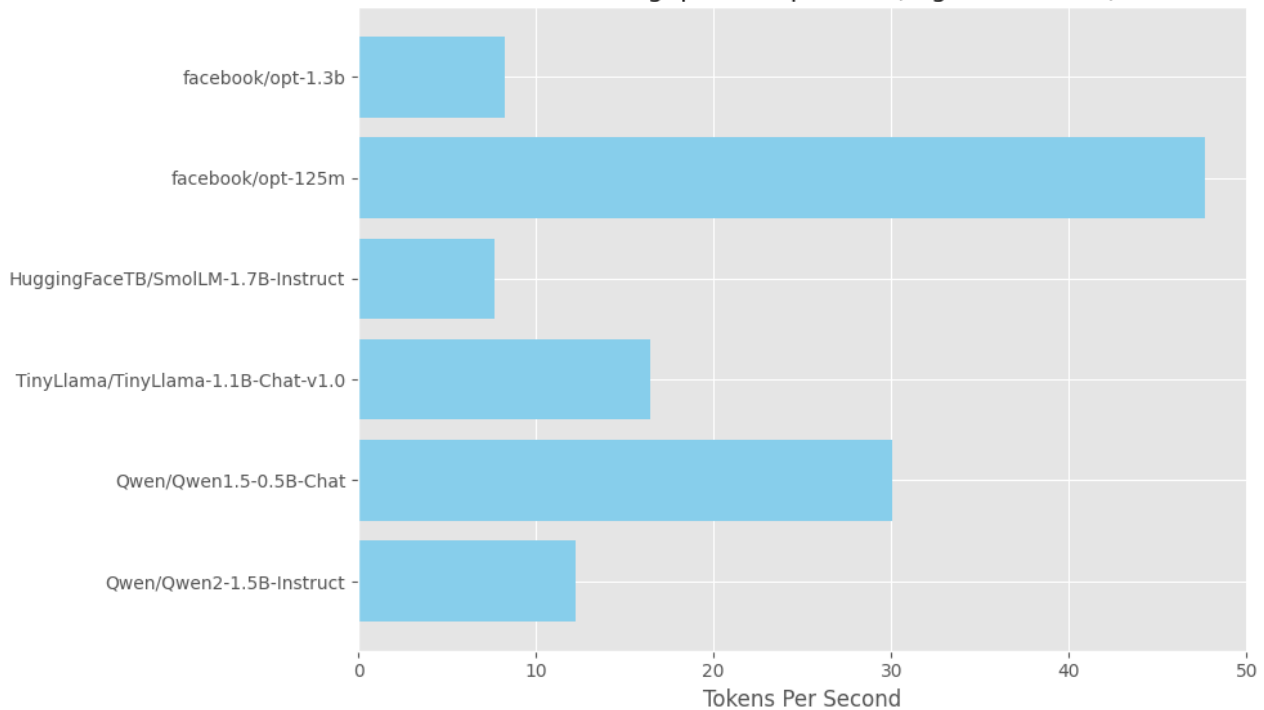
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|                   |       |       |      |
|-------------------|-------|-------|------|
| facebook/opt-125m | 2.85  | 47.64 | 0.37 |
| facebook/opt-1.3b | 16.51 | 8.27  | 0.52 |

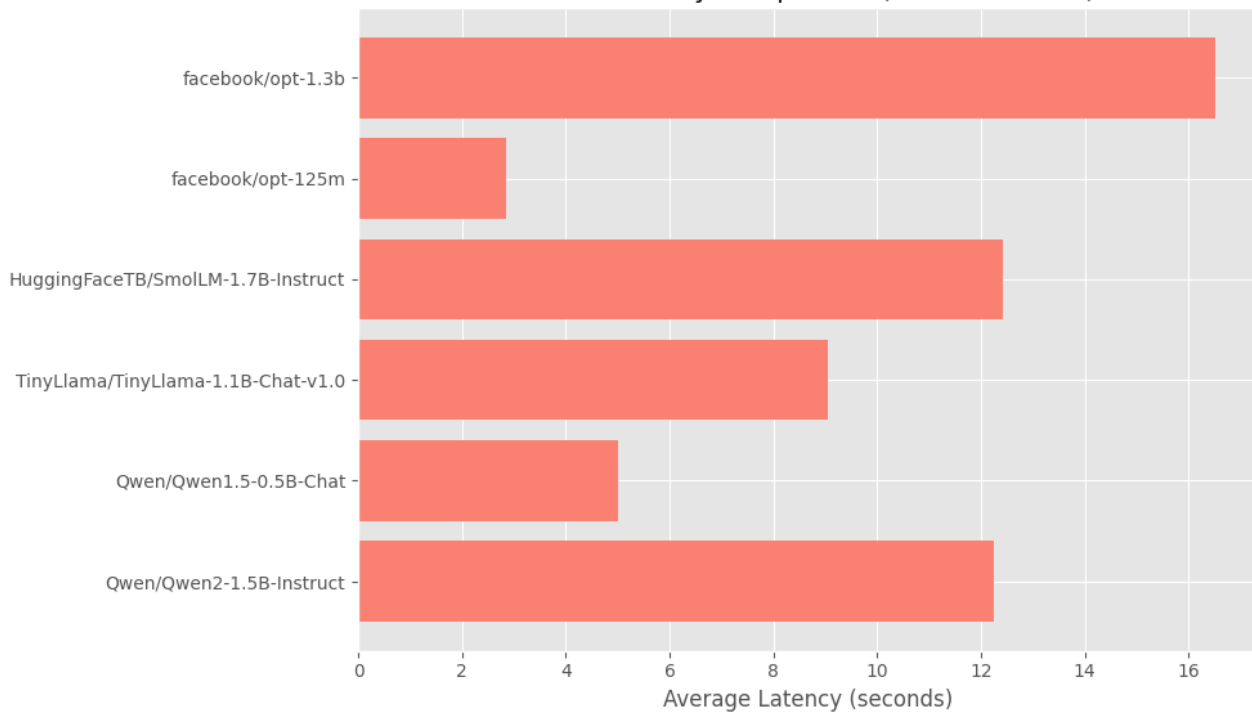
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## 4. Visual Analysis

Model Throughput Comparison (Higher is Better)



Model Latency Comparison (Lower is Better)



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